

MMI 804 · MATHEMATICAL SYSTEM MODELLING AND SIMULATION

MSc in Mathematical Innovation · Open University of Kenya

Topic 5: Deterministic vs Stochastic Models

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LEARNING OUTCOMES

1. Distinguish deterministic models from stochastic models and identify the appropriate choice for a given system.
2. Incorporate probability distributions into system state transitions using the Gillespie algorithm.
3. Implement Monte Carlo simulation in Python and interpret ensemble statistics.
4. Analyse the impact of demographic and environmental stochasticity on long-term system stability and extinction risk.

1 Topic Overview

A deterministic model, given identical initial conditions, always produces the same output. A stochastic model incorporates randomness — different realisations of the same model will produce different trajectories, even from identical starting points. Both are useful; the choice depends on whether randomness is a fundamental feature of the system, or a negligible complication.

WHEN DOES RANDOMNESS MATTER?

Randomness matters most when: (1) population sizes are small (discrete stochastic effects dominate); (2) rare events (disease introduction, mutation, fire ignition) are critical; (3) the system is near a threshold or bifurcation; (4) we need to characterise risk distributions (not just mean outcomes). For large, homogeneous populations far from thresholds, deterministic models often provide excellent approximations.

2 Deterministic Models

A deterministic model is specified by:

$$\frac{d\mathbf{x}}{dt} = f(\mathbf{x}, t, \boldsymbol{\theta}), \quad \mathbf{x}(0) = \mathbf{x}_0$$

Given $\mathbf{x}_0$ and $\boldsymbol{\theta}$, the trajectory $\mathbf{x}(t)$ is uniquely determined. Key properties:

- **Reproducibility:** Same input $\implies$ same output.

- **Tractability:** Equilibria, stability, and bifurcations can be studied analytically.
- **Limitation:** Cannot capture intrinsic randomness or variance around predictions.

3 Sources of Stochasticity

TWO TYPES OF STOCHASTICITY

Demographic stochasticity: Arises from the discrete, probabilistic nature of individual-level events (birth, death, infection) in finite populations. Even if the average birth rate is r , each individual either does or does not reproduce in a given time step. This form of randomness is *intrinsic* to the system and cannot be reduced by improving knowledge. It is most important at small population sizes.

Environmental stochasticity: Arises from unpredictable variation in external conditions (rainfall, temperature, food availability) that affect all individuals simultaneously. This form of randomness is *extrinsic* — driven by forces outside the model boundary. It is important at all population sizes.

4 Stochastic Differential Equations (SDEs)

A continuous-time stochastic model is written as:

$$d\mathbf{x} = f(\mathbf{x}, t) dt + \sigma(\mathbf{x}, t) dW_t$$

where dW_t is the increment of a Wiener process (standard Brownian motion): $dW_t \sim \mathcal{N}(0, dt)$.

EXAMPLE 1: Stochastic Logistic Growth

Deterministic model: $\frac{dN}{dt} = rN \left(1 - \frac{N}{K}\right)$

Stochastic (SDE) version: $dN = rN \left(1 - \frac{N}{K}\right) dt + \sigma N dW_t$

The noise term $\sigma N dW_t$ represents environmental stochasticity: the growth rate fluctuates randomly around r . The multiplicative form (σN) ensures variance scales with population size.

Python (Euler-Maruyama scheme):

```
import numpy as np
import matplotlib.pyplot as plt

def stochastic_logistic(r=0.3, K=1000, sigma=0.05,
                      N0=50, dt=0.1, T=100, n_sims=100):
    t = np.arange(0, T, dt)
```

```

all_paths = []
for _ in range(n_sims):
    N = np.zeros(len(t)); N[0] = N0
    for i in range(1, len(t)):
        dW = np.random.normal(0, np.sqrt(dt))
        N[i] = N[i-1] + r*N[i-1]*(1-N[i-1]/K)*dt \
            + sigma*N[i-1]*dW
        N[i] = max(N[i], 0) # absorbing boundary at 0
    all_paths.append(N)
return t, np.array(all_paths)

t, paths = stochastic_logistic()
plt.plot(t, paths[:20].T, alpha=0.3, color='teal')
plt.plot(t, np.mean(paths, axis=0), 'k-', lw=2, label='Mean')
plt.xlabel('Time'); plt.ylabel('N'); plt.legend()

```

5 The Gillespie Algorithm (Stochastic Simulation Algorithm)

For small populations, continuous approximations break down. The **Gillespie Algorithm** (1977) provides an *exact* stochastic simulation of a system of reactions, treating events (births, deaths, infections) as probabilistic processes.

GILLESPIE ALGORITHM — Steps

Given a system with M possible event types, each with rate (propensity) $a_j(\mathbf{x})$:

1. Compute total propensity: $A = \sum_{j=1}^M a_j(\mathbf{x})$.
2. Sample time to next event: $\tau \sim \text{Exponential}(A)$, i.e., $\tau = -\ln(u_1)/A$ where $u_1 \sim \text{Uniform}(0, 1)$.
3. Select which event occurs: choose event j where $\sum_{k=1}^{j-1} a_k < u_2 A \leq \sum_{k=1}^j a_k$, with $u_2 \sim \text{Uniform}(0, 1)$.
4. Update state $\mathbf{x}$ according to the chosen event and advance time by τ .
5. Repeat until the end time.

EXAMPLE 2: Stochastic SIR via Gillespie

Events and propensities:

$$\text{Infection: } S \rightarrow I \quad \text{rate } a_1 = \frac{\beta SI}{N}$$

$$\text{Recovery: } I \rightarrow R \quad \text{rate } a_2 = \gamma I$$

```
def gillespie_SIR(beta, gamma, S0, I0, N, t_max):
```

```

S, I, R = S0, I0, N - S0 - I0
t, traj = 0, [(t, S, I, R)]
while t < t_max and I > 0:
    a1 = beta * S * I / N # infection
    a2 = gamma * I        # recovery
    A = a1 + a2
    if A == 0: break
    tau = -np.log(np.random.uniform()) / A
    t += tau
    if np.random.uniform() < a1 / A:
        S -= 1; I += 1 # infection event
    else:
        I -= 1; R += 1 # recovery event
    traj.append((t, S, I, R))
return np.array(traj)

```

For small populations (e.g., $N = 50$), the Gillespie trajectories show significant variation — including stochastic extinction even when $\mathcal{R}_0 > 1$.

6 Monte Carlo Simulation

Monte Carlo methods use repeated random sampling to compute quantities that are analytically intractable. In system modelling, they are used to:

- Propagate *parameter uncertainty* through a model (uncertainty quantification).
- Estimate the distribution of model outputs when inputs are uncertain.
- Compute integrals by sampling.

EXAMPLE 3: Uncertainty in $\mathcal{R}_0$ — Monte Carlo

Scenario: β and γ are known with uncertainty: $\beta \sim \mathcal{N}(0.3, 0.03^2)$, $\gamma \sim \mathcal{N}(0.1, 0.01^2)$.

```

n_samples = 10000
beta_samples = np.random.normal(0.3, 0.03, n_samples)
gamma_samples = np.random.normal(0.1, 0.01, n_samples)
R0_samples = beta_samples / gamma_samples

print(f"Mean R0: {R0_samples.mean():.2f}")
print(f"95% CI: [{np.percentile(R0_samples, 2.5):.2f},
                {np.percentile(R0_samples, 97.5):.2f}]",)
print(f"P(R0 > 1): {(R0_samples > 1).mean():.3f}")

```

7 Extinction Risk and Stochastic Thresholds

One of the most important applications of stochastic modelling is assessing **extinction risk** — the probability that a population (or epidemic) dies out due to random fluctuations, even when the deterministic model predicts persistence.

EXAMPLE 4: Extinction Probability for Small Populations

Branching process approximation: For an SIR-type epidemic with $\mathcal{R}_0 > 1$ but starting with I_0 infected individuals, the probability of stochastic extinction before epidemic takeoff is approximately:

$$P(\text{extinction}) \approx \left(\frac{1}{\mathcal{R}_0} \right)^{I_0}$$

Example: $\mathcal{R}_0 = 2.5$, $I_0 = 3$: $P(\text{extinction}) \approx (0.4)^3 = 0.064$.

So even with $\mathcal{R}_0 = 2.5$, there is a 6.4% chance the epidemic self-extinguishes before taking off — a result invisible to the deterministic SIR model.

8 Key Concepts and Terminology

Deterministic Model	Given identical inputs, always produces identical outputs.
Stochastic Model	Incorporates randomness; different realisations produce different trajectories.
Demographic Stochasticity	Intrinsic randomness from discrete probabilistic individual events; important at small N .
Environmental Stochasticity	Extrinsic randomness from fluctuating environmental conditions; important at all N .
SDE	Stochastic Differential Equation: ODE with added noise term driven by Wiener process.
Gillespie Algorithm	Exact stochastic simulation algorithm for systems of probabilistic events.
Monte Carlo	Method of computing quantities via repeated random sampling.
Wiener Process	Mathematical model of Brownian motion: W_t with $dW_t \sim \mathcal{N}(0, dt)$.
Extinction Risk	Probability that a stochastic population falls to zero despite deterministic persistence.

9 Revision Questions

1. Run 500 realisations of the stochastic logistic model (Example 1) with $N_0 = 10$, $K = 1000$, $\sigma = 0.1$. What fraction goes extinct before $t = 50$? Compare with $N_0 = 100$. What does this reveal?
2. Implement the Gillespie SIR algorithm (Example 2) for a school of $N = 300$ students with $\mathcal{R}_0 = 2$. Run 200 realisations starting with $I_0 = 1$. Report: mean epidemic size, fraction of realisations with < 5 total cases (stochastic extinction), and 95% confidence band for $I(t)$.
3. Derive the Euler-Maruyama update rule for the SDE $dN = f(N)dt + g(N)dW_t$ by discretising over a small time interval Δt .
4. For the logistic SDE with multiplicative noise, compute the expected value $E[N(t)]$ and variance $\text{Var}[N(t)]$ at steady state (hint: use Itô's lemma or numerical Monte Carlo).
5. Explain why the Gillespie Algorithm produces exact samples from the stochastic process, while Euler-Maruyama only approximates the SDE.
6. A conservation manager wants to know the probability that a rhino population of $N = 20$ goes extinct within 50 years. Set up a stochastic birth-death model with birth rate $\lambda = 0.15$ per individual per year and death rate $\mu = 0.12$ per individual per year. Run Monte Carlo simulations and report the extinction probability.
7. Compare deterministic and stochastic SIR model predictions for a measles outbreak with $N = 500$, $\mathcal{R}_0 = 15$, starting with $I_0 = 1$. At what population size do the two models agree closely?

TOPIC 5 SUMMARY

- Deterministic models predict unique trajectories; stochastic models generate distributions of outcomes — the choice depends on whether randomness is intrinsically important.
- Demographic stochasticity dominates at small N ; environmental stochasticity affects all scales.
- SDEs extend ODEs by adding noise terms; the Euler-Maruyama method provides a simple numerical integrator.
- The Gillespie Algorithm provides exact stochastic simulation for systems with discrete probabilistic events — essential for small populations.
- Monte Carlo simulation quantifies uncertainty in model outputs arising from uncertain parameters.
- *The deterministic model tells you where the system tends to go; the stochastic model tells you how likely it is to get there.*

10 Further Reading

- **Gillespie, D.T.** (1977). Exact stochastic simulation of coupled chemical reactions. *Journal of Physical Chemistry*, 81(25), 2340–2361.

- **Allen, L.J.S.** (2010). *An Introduction to Stochastic Processes with Applications to Biology* (2nd ed.). CRC Press.
- **Oksendal, B.** (2013). *Stochastic Differential Equations* (6th ed.). Springer.